

The Art of T_EX

S. VanDenDriessche

Department of Mathematics
University of Notre Dame / University of Saint Francis

VERUM, May 30, 2013

What *is* T_EX?

- T_EX and L^AT_EX (pronounced “tech” and “lay-tech”) are programming languages that are the centers of a collection of programs used for typesetting, particularly for scientific documents.
- In common usage the terms “T_EX” and “L^AT_EX” are often interchangeable and can denote a collection of programs (i.e. “I am going to go T_EX up my homework”).
- Note that T_EX is *not* a word-processing program, as the data input is different than the final output.

What *is* T_EX?

- T_EX and L^AT_EX (pronounced “tech” and “lay-tech”) are programming languages that are the centers of a collection of programs used for typesetting, particularly for scientific documents.
- In common usage the terms “T_EX” and “L^AT_EX” are often interchangeable and can denote a collection of programs (i.e. “I am going to go T_EX up my homework”).
- Note that T_EX is *not* a word-processing program, as the data input is different than the final output.

What *is* T_EX?

- T_EX and L^AT_EX (pronounced “tech” and “lay-tech”) are programming languages that are the centers of a collection of programs used for typesetting, particularly for scientific documents.
- In common usage the terms “T_EX” and “L^AT_EX” are often interchangeable and can denote a collection of programs (i.e. “I am going to go T_EX up my homework”).
- Note that T_EX is *not* a word-processing program, as the data input is different than the final output.

Some History

We can thank Donald E. Knuth (the creator of the surreal numbers) for these awesome typesetting programs.



Some History

- 1978 - Knuth becomes frustrated with his publisher's typesetting and resolves to develop his own.
- 1979 - $\text{T}_{\text{E}}\text{X}$ is introduced to the world at an AMS meeting and mathematicians frenzy.
- 1986 - Leslie Lamport introduces $\text{\LaTeX}$, a higher level language with all the power of $\text{T}_{\text{E}}\text{X}$.
- 1989 - $\text{T}_{\text{E}}\text{X}$ language is frozen, and officially complete.
- 2013 - Everyone at VERUM learns $\text{T}_{\text{E}}\text{X}$ and loves it!

Some History

- 1978 - Knuth becomes frustrated with his publisher's typesetting and resolves to develop his own.
- 1979 - $\text{T}_{\text{E}}\text{X}$ is introduced to the world at an AMS meeting and mathematicians frenzy.
- 1986 - Leslie Lamport introduces $\text{\LaTeX}$, a higher level language with all the power of $\text{T}_{\text{E}}\text{X}$.
- 1989 - $\text{T}_{\text{E}}\text{X}$ language is frozen, and officially complete.
- 2013 - Everyone at VERUM learns $\text{T}_{\text{E}}\text{X}$ and loves it!

Some History

- 1978 - Knuth becomes frustrated with his publisher's typesetting and resolves to develop his own.
- 1979 - $\text{T}_{\text{E}}\text{X}$ is introduced to the world at an AMS meeting and mathematicians frenzy.
- 1986 - Leslie Lamport introduces $\text{\LaTeX}$, a higher level language with all the power of $\text{T}_{\text{E}}\text{X}$.
- 1989 - $\text{T}_{\text{E}}\text{X}$ language is frozen, and officially complete.
- 2013 - Everyone at VERUM learns $\text{T}_{\text{E}}\text{X}$ and loves it!

Some History

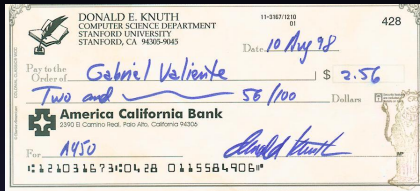
- 1978 - Knuth becomes frustrated with his publisher's typesetting and resolves to develop his own.
- 1979 - $\text{T}_{\text{E}}\text{X}$ is introduced to the world at an AMS meeting and mathematicians frenzy.
- 1986 - Leslie Lamport introduces $\text{\LaTeX}$, a higher level language with all the power of $\text{T}_{\text{E}}\text{X}$.
- 1989 - $\text{T}_{\text{E}}\text{X}$ language is frozen, and officially complete.
- 2013 - Everyone at VERUM learns $\text{T}_{\text{E}}\text{X}$ and loves it!

Some History

- 1978 - Knuth becomes frustrated with his publisher's typesetting and resolves to develop his own.
- 1979 - $\text{T}_{\text{E}}\text{X}$ is introduced to the world at an AMS meeting and mathematicians frenzy.
- 1986 - Leslie Lamport introduces $\text{\LaTeX}$, a higher level language with all the power of $\text{T}_{\text{E}}\text{X}$.
- 1989 - $\text{T}_{\text{E}}\text{X}$ language is frozen, and officially complete.
- 2013 - Everyone at VERUM learns $\text{T}_{\text{E}}\text{X}$ and loves it!

Motivations for using L^AT_EX

- Checks from Knuth!

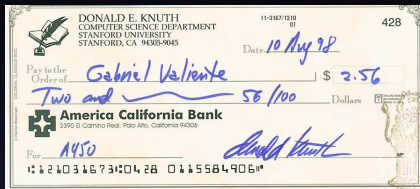


- T_EX has the best-aligned, most visually pleasing output of any accessible typesetter.
The flow of writing is T_EX is “natural” and flows without too much thought (font selection, size selection, etc.).
- Without T_EX, the following joke would be very difficult to

$$\text{If } \lim_{x \rightarrow 8} \frac{1}{x-8} = \infty \text{ then } \lim_{x \rightarrow 5} \frac{1}{x-5} = \infty$$

Motivations for using L^AT_EX

- Checks from Knuth!



- T_EX has the best-aligned, most visually pleasing output of any accessible typesetter.

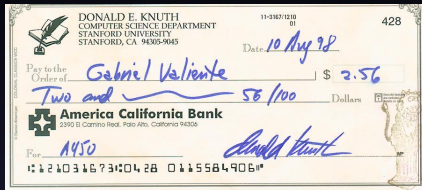
The flow of writing in T_EX is “natural” and flows without too much thought (font selection, size selection, etc.).

- Without T_EX, the following joke would be very difficult to

$$\text{If } \lim_{x \rightarrow 8} \frac{1}{x-8} = \infty \text{ then } \lim_{x \rightarrow 5} \frac{1}{x-5} = \omega$$

Motivations for using L^AT_EX

- Checks from Knuth!

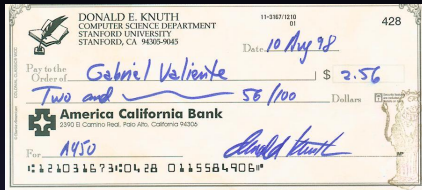


- T_EX has the best-aligned, most visually pleasing output of any accessible typesetter.
- The flow of writing is T_EX is “natural” and flows without too much thought (font selection, size selection, etc.).
- Without T_EX, the following joke would be very difficult to

$$\text{If } \lim_{x \rightarrow 8} \frac{1}{x-8} = \infty \text{ then } \lim_{x \rightarrow 5} \frac{1}{x-5} = \infty$$

Motivations for using L^AT_EX

- Checks from Knuth!

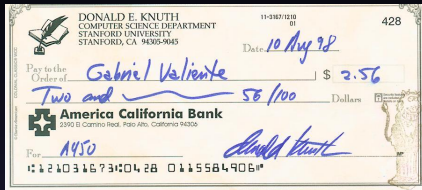


- T_EX has the best-aligned, most visually pleasing output of any accessible typesetter.
- The flow of writing in T_EX is “natural” and flows without too much thought (font selection, size selection, etc.).
- Without T_EX, the following joke would be very difficult to typeset:

$$\text{If } \lim_{x \rightarrow 8} \frac{1}{x-8} = \infty \text{ then } \lim_{x \rightarrow 5} \frac{1}{x-5} = \omega$$

Motivations for using L^AT_EX

- Checks from Knuth!



- T_EX has the best-aligned, most visually pleasing output of any accessible typesetter.
- The flow of writing in T_EX is “natural” and flows without too much thought (font selection, size selection, etc.).
- Without T_EX, the following joke would be very difficult to typeset:

$$\text{If } \lim_{x \rightarrow 8} \frac{1}{x-8} = \infty \text{ then } \lim_{x \rightarrow 5} \frac{1}{x-5} = \infty$$

More Motivations (from CTAN)

- $\text{\TeX}$ is stable and will never eat your document.
- Ease and power of $\text{\TeX}$'s macros make for a flexible and adaptable typesetting program.
- The input is a plain text file - entire books can be sent as an email attachment!
- $\text{\TeX}$ is free!
- $\text{\TeX}$ is the standard format for paper submission, research presentation (beamer), poster session presentation, etc.

More Motivations (from CTAN)

- T_EX is stable and will never eat your document.
- Ease and power of T_EX 's macros make for a flexible and adaptable typesetting program.
- The input is a plain text file - entire books can be sent as an email attachment!
- T_EX is free!
- T_EX is the standard format for paper submission, research presentation (beamer), poster session presentation, etc.

More Motivations (from CTAN)

- T_EX is stable and will never eat your document.
- Ease and power of T_EX 's macros make for a flexible and adaptable typesetting program.
- The input is a plain text file - entire books can be sent as an email attachment!
- T_EX is free!
- T_EX is the standard format for paper submission, research presentation (beamer), poster session presentation, etc.

More Motivations (from CTAN)

- T_EX is stable and will never eat your document.
- Ease and power of T_EX 's macros make for a flexible and adaptable typesetting program.
- The input is a plain text file - entire books can be sent as an email attachment!
- T_EX is free!
- T_EX is the standard format for paper submission, research presentation (beamer), poster session presentation, etc.

More Motivations (from CTAN)

- T_EX is stable and will never eat your document.
- Ease and power of T_EX 's macros make for a flexible and adaptable typesetting program.
- The input is a plain text file - entire books can be sent as an email attachment!
- T_EX is free!
- T_EX is the standard format for paper submission, research presentation (beamer), poster session presentation, etc.

The Main Result

Theorem (Knuth)

The learning curve of $\text{\LaTeX}$ is very steep.

Corollary

Stick with it and you will be fine.

The Main Result

Theorem (Knuth)

The learning curve of $\text{\LaTeX}$ is very steep.

Corollary

Stick with it and you will be fine.

The Main Result

Theorem (Knuth)

The learning curve of $\text{\LaTeX}$ is very steep.

Corollary

Stick with it and you will be fine.

What do we need to start?

The first steps into the world of $\text{T}_{\text{E}}\text{X}$ consist of downloading a version of $\text{T}_{\text{E}}\text{X}$ and an editor.

- Miktex is really the best version of $\text{T}_{\text{E}}\text{X}$ to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - $\text{\LaTeX}$ Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning $\text{T}_{\text{E}}\text{X}$.

What do we need to start?

The first steps into the world of $\text{T}_{\text{E}}\text{X}$ consist of downloading a version of $\text{T}_{\text{E}}\text{X}$ and an editor.

- Miktex is really the best version of $\text{T}_{\text{E}}\text{X}$ to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - $\text{\LaTeX}$ Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning $\text{T}_{\text{E}}\text{X}$.

What do we need to start?

The first steps into the world of $\text{T}_{\text{E}}\text{X}$ consist of downloading a version of $\text{T}_{\text{E}}\text{X}$ and an editor.

- Miktex is really the best version of $\text{T}_{\text{E}}\text{X}$ to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - $\text{\LaTeX}$ Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning $\text{T}_{\text{E}}\text{X}$.

What do we need to start?

The first steps into the world of $\text{T}_{\text{E}}\text{X}$ consist of downloading a version of $\text{T}_{\text{E}}\text{X}$ and an editor.

- Miktex is really the best version of $\text{T}_{\text{E}}\text{X}$ to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - $\text{\LaTeX}$ Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning $\text{T}_{\text{E}}\text{X}$.

What do we need to start?

The first steps into the world of T_EX consist of downloading a version of T_EX and an editor.

- Miktex is really the best version of T_EX to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - L^AT_EX Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning T_EX.

What do we need to start?

The first steps into the world of $\text{T}_{\text{E}}\text{X}$ consist of downloading a version of $\text{T}_{\text{E}}\text{X}$ and an editor.

- Miktex is really the best version of $\text{T}_{\text{E}}\text{X}$ to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - $\text{\LaTeX}$ Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning $\text{T}_{\text{E}}\text{X}$.

What do we need to start?

The first steps into the world of $\text{T}_{\text{E}}\text{X}$ consist of downloading a version of $\text{T}_{\text{E}}\text{X}$ and an editor.

- Miktex is really the best version of $\text{T}_{\text{E}}\text{X}$ to download and can be found by googling “Miktex.”
- Editor choices vary. Some good options include:
 - the editor included on MACs,
 - TexMaker (my personal favorite)
 - $\text{\LaTeX}$ Editor (LED).

Downloading these programs and getting them to interact is the most difficult aspect of learning $\text{T}_{\text{E}}\text{X}$.

The Preamble

Just like in other programming contexts, most input files need a preamble to tell T_EX what it is that you are doing. Preambles consist of the following:

- document class (article, letter, etc.),
- packages used (amssymb, amsmath, graphicsx, etc.)
- new commands, macros.

The Preamble

Just like in other programming contexts, most input files need a preamble to tell T_EX what it is that you are doing. Preambles consist of the following:

- document class (article, letter, etc.),
- packages used (amssymb, amsmath, graphicsx, etc.)
- new commands, macros.

The Preamble

Just like in other programming contexts, most input files need a preamble to tell T_EX what it is that you are doing. Preambles consist of the following:

- document class (article, letter, etc.),
- packages used (amssymb, amsmath, graphicsx, etc.)
- new commands, macros.

The Preamble

Just like in other programming contexts, most input files need a preamble to tell T_EX what it is that you are doing. Preambles consist of the following:

- document class (article, letter, etc.),
- packages used (amssymb, amsmath, graphicsx, etc.)
- new commands, macros.

The Preamble

Lets write a simple preamble to prepare for some future exercises:

```
\documentclass{article}  
\usepackage{amsmath}  
\usepackage{amsfonts}  
\usepackage{amssymb}
```

The Preamble

Lets write a simple preamble to prepare for some future exercises:

```
\documentclass{article}  
\usepackage{amsmath}  
\usepackage{amsfonts}  
\usepackage{amssymb}
```

Getting Started

Try inputting and compiling the following text.

```
This is some text! Woot!           What  
is      so      interesting  
about   this?
```

When I compile this, I get:

```
This is some text! Woot! What is so interesting about this?
```

fact:

Repeated white spaces are treated by $T_E X$ as a single white space.

Getting Started

Try inputting and compiling the following text.

```
This is some text! Woot!           What  
is      so      interesting  
about   this?
```

When I compile this, I get:

```
This is some text! Woot! What is so interesting about this?
```

Wait!

Repeated white spaces are treated by $T_E X$ as a single white space.

Getting Started

Try inputting and compiling the following text.

```
This is some text! Woot!           What  
is      so      interesting  
about   this?
```

When I compile this, I get:

```
This is some text! Woot! What is so interesting about this?
```

Wait!

Repeated white spaces are treated by $T_E X$ as a single white space.

Getting Started

Try inputting and compiling the following text.

```
This is some text! Woot!           What  
is      so      interesting  
about   this?
```

When I compile this, I get:

```
This is some text! Woot! What is so interesting about this?
```

Fact

Repeated white spaces are treated by $T_E X$ as a single white space.

More Formatting

Linebreaks are handled similarly by $\text{T}_{\text{E}}\text{X}$ (repeating does nothing, might not even compile). To break a line, type

```
\\
```

Add linebreaks to our previously existing text to get

```
This is some text!\\ Woot!      \\      What  
is          so      interesting  
about      this?
```

This is some text!

Woot!

What is so interesting about this?

More Formatting

Linebreaks are handled similarly by $\text{T}_{\text{E}}\text{X}$ (repeating does nothing, might not even compile). To break a line, type

```
\\
```

Add linebreaks to our previously existing text to get

```
This is some text!\\ Woot!      \\      What  
is          so          interesting  
about      this?
```

This is some text!
Woot!

What is so interesting
about this?

More Formatting

Linebreaks are handled similarly by T_EX(repeating does nothing, might not even compile). To break a line, type

```
\\
```

Add linebreaks to our previously existing text to get

```
This is some text!\\ Woot!      \\      What  
is          so          interesting  
about      this?
```

This is some text!

Woot!

What is so interesting about this?

Commands

What do you think of when you think of “command?”

Commands



Do as I command!!!

Commands

Commands do exactly what you think - they tell $\text{\LaTeX}$ to do something - other than the text of the command. To distinguish from regular text, commands are signaled by

```
\
```

Some commands have an input, while others just tell $\text{\TeX}$ to do something.

```
\emph{This text will be emphasized}.
```

This text will be emphasized.

Commands

Commands do exactly what you think - they tell $\text{\LaTeX}$ to do something - other than the text of the command. To distinguish from regular text, commands are signaled by

```
\
```

Some commands have an input, while others just tell $\text{\TeX}$ to do something.

```
\emph{This text will be emphasized}.
```

This text will be emphasized.

Commands

Commands do exactly what you think - they tell $\text{\LaTeX}$ to do something - other than the text of the command. To distinguish from regular text, commands are signaled by

```
\
```

Some commands have an input, while others just tell $\text{\TeX}$ to do something.

```
\emph{This text will be emphasized}.
```

This text will be emphasized.

Commands

Commands do exactly what you think - they tell $\text{\LaTeX}$ to do something - other than the text of the command. To distinguish from regular text, commands are signaled by

```
\
```

Some commands have an input, while others just tell $\text{\TeX}$ to do something.

```
\emph{This text will be emphasized}.
```

This text will be emphasized.

Environments

Environments can be used in $\text{\LaTeX}$ to format the input text in specific ways, such as alignments, lists, colors, and so on. We must use commands to tell $\text{\TeX}$ that we are entering or exiting an environment.

```
\begin{Large}  
This text should be bigger.  
\end{Large}
```

This text should be bigger.

In general, environments usually are inputted in the form

```
\begin{environment}[options]  
  
\end{environment}
```

Environments

Environments can be used in $\text{\LaTeX}$ to format the input text in specific ways, such as alignments, lists, colors, and so on. We must use commands to tell $\text{\TeX}$ that we are entering or exiting an environment.

```
\begin{Large}  
This text should be bigger.  
\end{Large}
```

This text should be bigger.

In general, environments usually are inputted in the form

```
\begin{environment}[options]  
  
\end{environment}
```

Environments

Environments can be used in $\text{\LaTeX}$ to format the input text in specific ways, such as alignments, lists, colors, and so on. We must use commands to tell $\text{\TeX}$ that we are entering or exiting an environment.

```
\begin{Large}  
This text should be bigger.  
\end{Large}
```

This text should be bigger.

In general, environments usually are inputted in the form

```
\begin{environment}[options]  
  
\end{environment}
```

Environments

Environments can be used in $\text{\LaTeX}$ to format the input text in specific ways, such as alignments, lists, colors, and so on. We must use commands to tell $\text{\TeX}$ that we are entering or exiting an environment.

```
\begin{Large}  
This text should be bigger.  
\end{Large}
```

This text should be bigger.

In general, environments usually are inputted in the form

```
\begin{environment}[options]  
text goes here.  
\end{environment}
```

Nesting Environments

Environments can be nested.

```
\begin{Large}  
\begin{quote}  
This text should be outputted in italics.  
\end{quote}  
\end{Large}
```

This text should be outputted in italics.

Nesting Environments

Environments can be nested.

```
\begin{Large}  
\begin{quote}  
This text should be outputted in italics.  
\end{quote}  
\end{Large}
```

This text should be outputted in italics.

Nesting Environments

Environments can be nested.

```
\begin{Large}  
\begin{quote}  
This text should be outputted in italics.  
\end{quote}  
\end{Large}
```

This text should be outputted in italics.

Itemizing

$\text{\LaTeX}$ is very good for bulleted lists.

```
\begin{itemize}
\item This is item 1.
\item This is item 2.
\end{itemize}
```

- This is item 1.
- This is item 2.

Itemizing

$\text{\LaTeX}$ is very good for bulleted lists.

```
\begin{itemize}  
\item This is item 1.  
\item This is item 2.  
\end{itemize}
```

- This is item 1.
- This is item 2.

Itemizing

$\text{\LaTeX}$ is very good for bulleted lists.

```
\begin{itemize}  
\item This is item 1.  
\item This is item 2.  
\end{itemize}
```

- This is item 1.
- This is item 2.

Itemizing

The *itemize* environment also allows custom markers. Note that this is a use of [...], which usually denote options for a given command.

```
\begin{itemize}  
\item[i.] This is item 1.  
\item[ii.] This is item 2.  
\end{itemize}
```

- i. This is item 1.
- ii. This is item 2.

Itemizing

The *itemize* environment also allows custom markers. Note that this is a use of [...], which usually denote options for a given command.

```
\begin{itemize}  
\item[i.] This is item 1.  
\item[ii.] This is item 2.  
\end{itemize}
```

- i. This is item 1.
- ii. This is item 2.

Itemizing

The *itemize* environment also allows custom markers. Note that this is a use of [...], which usually denote options for a given command.

```
\begin{itemize}  
\item[i.] This is item 1.  
\item[ii.] This is item 2.  
\end{itemize}
```

- i. This is item 1.
- ii. This is item 2.

Itemizing

enumerate is another common listing environment.

```
\begin{enumerate}  
\item This is item 1.  
\item This is item 2.  
\end{enumerate}
```

- 1 This is item 1.
- 2 This is item 2.

Itemizing

enumerate is another common listing environment.

```
\begin{enumerate}  
\item This is item 1.  
\item This is item 2.  
\end{enumerate}
```

- 1 This is item 1.
- 2 This is item 2.

Itemizing

enumerate is another common listing environment.

```
\begin{enumerate}  
\item This is item 1.  
\item This is item 2.  
\end{enumerate}
```

- 1 This is item 1.
- 2 This is item 2.

Math Mode

Math mode is an extremely common environment in $\text{\LaTeX}$ documents - in fact proper formatting of formulae was one of Knuth's primary motivations for developing $\text{\TeX}$.

- Math mode is “on” for all text between a pair of dollar signs ($\$math\text{ moded text goes here}\$$).
- If you try to use a math command outside of math mode, it is likely that your file will not compile.

There are environments and commands that do not compile when used inside of math mode.

- In math mode, there are a ton of commands to call mathematical symbols, and any plain text is italicized and treated as variable symbols.

Math Mode

Math mode is an extremely common environment in $\text{\LaTeX}$ documents - in fact proper formatting of formulae was one of Knuth's primary motivations for developing $\text{\TeX}$.

- Math mode is “on” for all text between a pair of dollar signs ($\$math\text{ moded text goes here}\$$).
- If you try to use a math command outside of math mode, it is likely that your file will not compile.

There are environments and commands that do not compile when used inside of math mode.

- In math mode, there are a ton of commands to call mathematical symbols, and any plain text is italicized and treated as variable symbols.

Math Mode

Math mode is an extremely common environment in $\text{\LaTeX}$ documents - in fact proper formatting of formulae was one of Knuth's primary motivations for developing $\text{\TeX}$.

- Math mode is “on” for all text between a pair of dollar signs ($\$$ math moded text goes here $\$$).
- If you try to use a math command outside of math mode, it is likely that your file will not compile.

There are environments and commands that do not compile when used inside of math mode.

- In math mode, there are a ton of commands to call mathematical symbols, and any plain text is italicized and treated as variable symbols.

Math Mode

Math mode is an extremely common environment in $\text{\LaTeX}$ documents - in fact proper formatting of formulae was one of Knuth's primary motivations for developing $\text{\TeX}$.

- Math mode is “on” for all text between a pair of dollar signs ($\$$ math moded text goes here $\$$).
- If you try to use a math command outside of math mode, it is likely that your file will not compile.
- There are environments and commands that do not compile when used inside of math mode.
- In math mode, there are a ton of commands to call mathematical symbols, and on plain text is italicized and treated as outside symbols.

Math Mode

Math mode is an extremely common environment in $\text{\LaTeX}$ documents - in fact proper formatting of formulae was one of Knuth's primary motivations for developing $\text{\TeX}$.

- Math mode is “on” for all text between a pair of dollar signs ($\$math\text{ moded text goes here}\$$).
- If you try to use a math command outside of math mode, it is likely that your file will not compile.
- There are environments and commands that do not compile when used inside of math mode.
- In math mode, there are a ton of commands to call mathematical symbols, and any plain text is italicized and treated as variable symbols.

Math Commands

It is sometimes considered improper to start a sentence with a symbol such as π , Σ , or $\mathcal{O}$.

It is sometimes considered improper to start a sentence with a symbol such as π , Σ , or $\mathcal{O}$.

Math Commands

It is sometimes considered improper to start a sentence with a symbol such as π , Σ , or $\mathcal{O}$.

It is sometimes considered improper to start a sentence with a symbol such as π , Σ , or $\mathcal{O}$.

Super and Subscript

Superscript and subscript work as one might expect.

Sentences like `'$\forall x(x=x)$'` and `'$\exists x(\pi^x=2)$'` are Δ^0_2 .

Sentences like $\forall x(x = x)$ and $\exists x(\pi^x = 2)$ are Δ^0_2 .

Super and Subscript

Superscript and subscript work as one might expect.

Sentences like `'$\forall x(x=x)$'` and
`'$\exists x(\pi^x=2)$'` are Δ^0_2 .

Sentences like $\forall x(x = x)$ and $\exists x(\pi^x = 2)$ are Δ^0_2 .

Super and Subscript

Superscript and subscript work as one might expect.

Sentences like `'$\forall x(x=x)$'` and
`'$\exists x(\pi^x=2)$'` are Δ^0_2 .

Sentences like $\forall x(x = x)$ and $\exists x(\pi^x = 2)$ are Δ^0_2 .

Super and Subscript

Towers are more complicated.

The integral of x^{x^x} is hard.

The integral of x^{x^x} is hard.

Super and Subscript

Towers are more complicated.

The integral of x^{x^x} is hard.

The integral of x^{x^x} is hard.

Super and Subscript

Towers are more complicated.

The integral of x^{x^x} is hard.

The integral of x^{x^x} is hard.

Math Environments

There are environments that only work in math mode. Indeed, *displaymath* is a special environment.

```
\begin{displaymath}
K(x) = \left\{ \begin{array}{ll}
1 & \text{if } x \in W_x \\
0 & \text{otherwise}
\end{array} \right.
\end{displaymath}
```

$$K(x) = \begin{cases} 1 & \text{if } x \in W_x \\ 0 & \text{otherwise} \end{cases}$$

Math Environments

There are environments that only work in math mode. Indeed, *displaymath* is a special environment.

```
\begin{displaymath}
K(x) = \left\{ \begin{array}{ll}
1 & \text{if } x \in W_x \\
0 & \text{otherwise}
\end{array} \right.
\end{displaymath}
```

$$K(x) = \begin{cases} 1 & \text{if } x \in W_x \\ 0 & \text{otherwise} \end{cases}$$

Math Environments

There are environments that only work in math mode. Indeed, *displaymath* is a special environment.

```
\begin{displaymath}
K(x) = \left\{ \begin{array}{ll}
1 & \text{if } x \in W_x \\
0 & \text{otherwise}
\end{array} \right.
\end{displaymath}
```

$$K(x) = \begin{cases} 1 & \text{if } x \in W_x \\ 0 & \text{otherwise} \end{cases}$$

Displaymath

The *displaymath* environment can also be signalled by double dollar signs.

```
$$x=\frac{-b\pm \sqrt{b^2-4ac}}{2a}$$
```

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Displaymath

The *displaymath* environment can also be signalled by double dollar signs.

```
$$x=\frac{-b\pm \sqrt{b^2-4ac}}{2a}$$
```

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Tips

- The internet is the T_EX user's best friend. Almost any imaginable problem is addressed there.
- Be aware of quote - " is different from " .
- Templates are a useful way to learn a document class.
- The best way to learn T_EX is to practice a lot.

I am here as a resource for you, please don't hesitate to ask!

Tips

- The internet is the $\text{T}_{\text{E}}\text{X}$ user's best friend. Almost any imaginable problem is addressed there.
- Be aware of quote - “ is different from ”.
- Templates are a useful way to learn a document class.
- The best way to learn $\text{T}_{\text{E}}\text{X}$ is to practice a lot.

I am here as a resource for you, please don't hesitate to ask!

Tips

- The internet is the $\text{T}_{\text{E}}\text{X}$ user's best friend. Almost any imaginable problem is addressed there.
- Be aware of quote - “ is different from ”.
- Templates are a useful way to learn a document class.
- The best way to learn $\text{T}_{\text{E}}\text{X}$ is to practice a lot.

I am here as a resource for you, please don't hesitate to ask!

Tips

- The internet is the T_EX user's best friend. Almost any imaginable problem is addressed there.
- Be aware of quote - “ is different from ”.
- Templates are a useful way to learn a document class.
- The best way to learn T_EX is to practice a lot.

I am here as a resource for you, please don't hesitate to ask!

Tips

- The internet is the T_EX user's best friend. Almost any imaginable problem is addressed there.
- Be aware of quote - “ is different from ”.
- Templates are a useful way to learn a document class.
- The best way to learn T_EX is to practice a lot.
- I am here as a resource for you, please don't hesitate to ask!